

MSDS 452: Graphical, Network, and Causal Models

Assignment 3

February 18, 2026

Causal inference of transport type in daily commute

Yingyu Mao

Abstract

I tested causal inference methods to identify the underlying data generation process in a synthetic dataset examining factors that influence daily commute transport choices. Using both constraint-based (Peter-Clark) and score-based (Greedy Equivalent Search) causal discovery algorithms, I compared the directed acyclic graphs generated from observational data containing demographic variables (age, gender, education, occupation, residence) and transport mode. The GES algorithm produced a better model that withstood independence testing of all d-separations, while the PC model contained a false independence claim. Notably, both algorithms correctly identified occupation as independent of the causal structure, a finding subsequently verified through targeted dependence testing. The results demonstrate that score-based causal discovery methods can provide reliable initial models when domain knowledge is limited, offering urban planners a systematic approach to understanding the causal mechanisms behind transportation choices for more effective policy interventions.

Keywords: Causal Discovery, Causal Inference, Independence Test

1.0 Introduction

Traditional statistical learning focuses on identifying the correlations between predictor variables and the response variable(s). It can generate predictions for the response variable without understanding the cause and effect. However, revealing the causality between variables goes beyond simple correlation-based predictions and enables direct effect isolation, controlling confounder effect, and predicting intervention outcomes. Causal inference is the statistical process that determines causal relationships between variables that drive the data generation process. In this paper, I will show the use of causal inference methods to reveal the causal relationships between the variables in a dataset about people's daily commute.

Suppose an urban planning consultant is tasked to understand what causes people's choice of transport. Standard predictive modeling might show that people in big cities use the subway more often, but it doesn't necessarily explain if that's due to the city's infrastructure, the type of jobs available there, or the demographic makeup of the residents. To answer causality questions and to make predictable urban planning changes, the following observational data was collected to build the causal model: the age, gender, education, occupation, residence and the means of transport.

2.0 Literature review

In causal inference, the hierarchical flow from cause to effect is modeled as Directed Acyclic Graphs (DAG). The nodes represent variables and the directed edges represent the functions that map one variable to another. The model imposes constraints on the variables so that each variable is independent of its non-descendants given its parents (Markov conditions). The parent (P) to child (C) relationship between nodes expresses the dependency of their probability distribution ($P(P, C) = P(C|P)P(P)$). By applying the chain rule of probability, the joint probability (likelihood) of all variables can be calculated as the product of the independent Markov kernels. Like all other machine learning algorithms, causal inference aims to find the optimal DAG model that best fits the data (maximizes the likelihood if uniform prior of the graph is assumed or maximizes a posteriori if prior knowledge of graph is known).

There are two separate approaches to establish the causal model. First, a tentative DAG can be generated manually based on human intuition and domain knowledge. The causal relationships between variables induce independence or conditional independence for d-separations, the path between nodes are blocked by colliders (in $A \rightarrow B \leftarrow C$, A and C are separated) or by conditioning on the middle nodes of a chain ($A \rightarrow B \rightarrow C$) or a fork ($A \leftarrow B \rightarrow C$) (Pearl, Glymour and Jewell 2016). Independence tests are then used to show if these conditional independences truly exist in the data. The current model is either accepted or rejected (Ness 2025). Modifications can be made directly to edges of the

failed tests, and further independence tests are carried out for each new model until the irrefutable model is reached.

Another approach, called causal discovery, directly searches for the DAG that explains the data with little prior knowledge. These methods can be further categorized into constraint-based and score-based algorithms (for number of data points much larger than the number of observed variables, usually less than 30). The former (mostly Peter-Clark algorithm (Spirtes, Glymour and Scheines 2001), others are variants that handles latent variables (Spirtes, Meek and Richardson 1995)) utilizes dependence and conditional independence as constraints to infer the edges and their directions. Score-based algorithms seek to maximize a Bayesian scoring criterion that measures local goodness-of-fit which only depends on the conditional frequency table for one variable and its parents. Greedy Equivalent Search is a two-phase algorithm that searches through equivalent classes of DAG instead of DAG themselves, which drastically reduces the search space (Chickering 2002). Similar exact search method introduced by Silander and Myllymäki (2006) follows the steps of identifying most likely parents of each node and the best sink node, and reordering the sequences of causal chain to maximize a Bayesian scoring criterion.

3.0 Methods

3.1 Dataset

I used a synthetic dataset introduced in the Causal AI book (Ness 2025). It contains 500 incidences of people's age (A), gender (G), education (E), occupation (O), residence (R) and the means of transport (T). All variables are categorical (the values of each variable are listed below). T is the outcome, and the others are the predictors.

```
A: ['adult' 'young' 'old']
S: ['F' 'M']
E: ['high' 'uni'] high: high school, uni: university
O: ['emp' 'self'] emp: employed, self: self-employed
R: ['small' 'big']
T: ['train' 'car' 'other']
```

3.2 d-separations and independence test

The d-separations were calculated using `get_independencies()` function from the `pgmpy` module.

The appropriate independence test for categorical variables is chi-squared test. A p-value threshold of 0.05 was chosen to determine dependences between variables: $p < 0.05$, dependence exists and the model is refuted. No family-wise error correction was used.

3.3 Causal discovery algorithms

I tested the Peter-Clark (PC) algorithm and Greedy Equivalent Search (GES) algorithm implemented in `causal-learn` python module. The scoring criterion in GES for discrete data is Bayesian Dirichlet equivalent uniform (BDeu) for Dirichlet distribution with uniform prior on graph structure.

4.0 Results

Instead of building a model manually, I first used the PC and GES algorithms to discover the causal model based on the data (Figure 1). Both identified the collider shape $A \rightarrow E \leftarrow S$ and excluded O from the connected component of the graph. Even after imposing a prior knowledge of $E \rightarrow O$ (education determines occupation), PC still rejected the connection to O. In addition, PC inferred a collider $E \rightarrow R \leftarrow T$ from the data, while GES inferred a chain $E \rightarrow R \rightarrow T$.

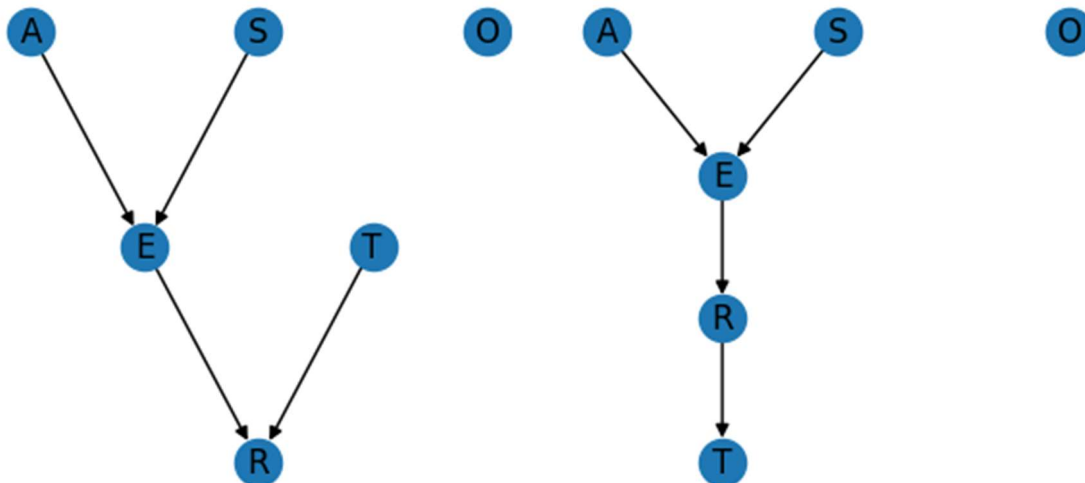


Figure 1 Causal models discovered by PC (left) and GES (right) algorithms.

Next, I looped through all d-separations and conducted chi-squared test of pair-wise independence on the two models generated by the causal discovery algorithms (Table 1). The GES model is irrefutable. The $T \perp A$ independence hypothesis in the PC model is false, that is the collider shape is false.

Table 1 Independence test result for all d-separations induced by PC and GES models.

PC model	GES model
$\{(R \perp S \mid A, E): \text{True}$ $(T \perp S \mid E, R): \text{True}$ $(T \perp A \mid E, R): \text{True}$ $(R \perp S \mid T, A, E): \text{True}$	$\{(T \perp E \mid R): \text{True}$ $(R \perp A \mid T, S, E): \text{True}$ $(T \perp S \mid A, R, E): \text{True}$ $(R \perp S \mid E): \text{True}$

(A ⊥ R S, E): True	(T ⊥ E A, R): True
(R ⊥ S T, E): True	(R ⊥ A S, E): True
(A ⊥ R T, E): True	(T ⊥ S R): True
(T ⊥ A S, E): True	(R ⊥ A T, E): True
(T ⊥ S): True	(T ⊥ S E): True
(A ⊥ S): True	(T ⊥ A S, E): True
(A ⊥ S T): True	(T ⊥ A E): True
(A ⊥ R E): True	(T ⊥ S A, R): True
(T ⊥ E A, S): True	(T ⊥ A R, E): True
(T ⊥ S E): True	(A ⊥ S): True
(T ⊥ A E): True	(T ⊥ S R, E): True
(R ⊥ S E): True	(T ⊥ A R, S): True
(T ⊥ E): True	(T ⊥ E R, S): True
(T ⊥ S A, E): True	(R ⊥ S A, E): True
(T ⊥ S A, E, R): True	(T ⊥ E A, R, S): True
(T ⊥ E S): True	(T ⊥ S A, E): True
(T ⊥ A S, E, R): True	(T ⊥ A R): True
(T ⊥ A S): True	(R ⊥ A E): True
(T ⊥ A): False	(R ⊥ S T, A, E): True
(T ⊥ S A): True	(R ⊥ S T, E): True
(A ⊥ R T, S, E): True	(T ⊥ A R, S, E): True}
(T ⊥ E A): True}	

To understand why both algorithms excluded O in the connected DAG, I then tested the dependence of each variable pair (results shown below). O is independent of all the other variables.

'A' and 'S' are independent.
'A' and 'E' are dependent.
'A' and 'O' are independent.
'A' and 'R' are dependent.
'A' and 'T' are dependent.
'S' and 'E' are dependent.
'S' and 'O' are independent.
'S' and 'R' are dependent.
'S' and 'T' are independent.
'E' and 'O' are independent.
'E' and 'R' are dependent.
'E' and 'T' are independent.
'O' and 'R' are independent.
'O' and 'T' are independent.
'R' and 'T' are dependent.

To see the effect of imposing a dependence of $E \rightarrow O$ in a causal model (Figure 2 left), I ran the independence test on its induced d-separations. All dependence pairs are conditioned on O (Figure 2 right), suggesting a false connection of O in the chain $E \rightarrow O \rightarrow R$. This verifies the result of causal discovery.

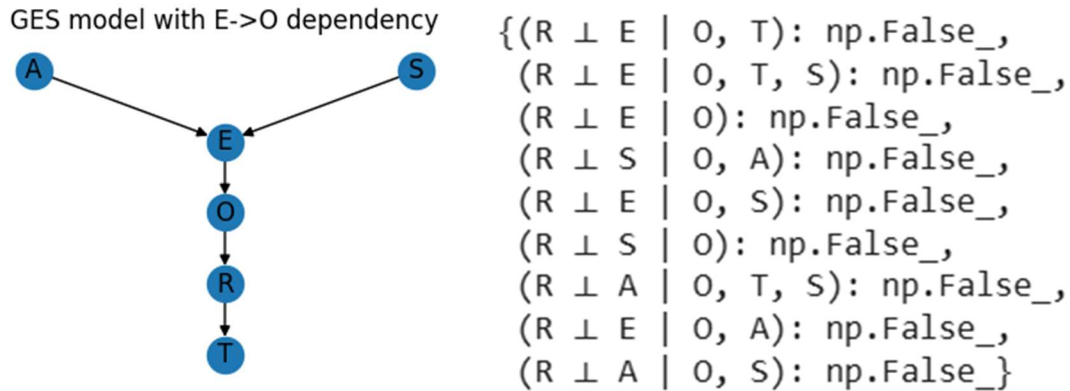


Figure 2 Imposing $E \rightarrow O$ dependency in the GES model.

5.0 Conclusion

Causal discovery provides a great starting point to uncover the DAG model when human knowledge in an area is limited or biased. The results show that the score-based GES algorithm constructed a better model than the constraint-based PC algorithm. The urban planning consultant can gather real-world data with more granularity (for instance, education and occupation can be more detailed. The categories used for occupation did not capture the dependencies between the two variables.), and expand predictor variable set to include residence type, parking availability and cost, commute distance, vehicle availability, train schedules, neighborhoods, etc. While it can be challenging to map out their dependencies manually, causal discovery algorithm in combination with independence tests to verify the discovered DAG is a viable approach. With more detailed and multi-dimensional descriptions of the demographic distribution, the urban planning consultant will be able to make more informed decisions on subway station or parking lot placement, etc.

References

- Chickering, D. M. 2002. "Optimal structure identification with greedy search." *Journal of machine learning research* 3, 507-554.
- Ness, Robert. 2025. "Causal AI." In *Chapter 4: Testing the DAG with causal constraints*. Shelter Island, NY: Manning.
- Pearl, Judea, Madelyn Glymour, and Nicholas P. Jewell. 2016. "Graphical Models and Their Applications." In *Causal Inference in Statistics: A Primer*, 35-52. West Sussex, United Kingdom: John Wiley & Sons Ltd.
- Silander, T., and P Myllymäki. 2006. "A Simple Approach for Finding the Globally Optimal Bayesian Network Structure." *Proceedings of the Twenty-Second Conference on Uncertainty in Artificial Intelligence*. 445-452.
- Spirtes, P., C. Meek, and T. Richardson. 1995. "Causal inference in the presence of latent variables and selection bias." *Proceedings of the Eleventh conference on Uncertainty in artificial intelligence* 499-506.
- Spirtes, Peter, Clark Glymour, and Richard Scheines. 2001. *Causation, Prediction, and Search*. MIT Press.